BuildLinka

Suggestions for real-world launch

What is needed to turn the prototype into an app that real vendors, workers, engineers and buyers can trust with real orders and real money, and how to test it with real people **before** building the backend. 1 October 2026.

Summary. The prototype is now much more complete: every category has a working calculator, search has barcode scanning, the engineer area works end to end, and the main forms are fixed. The next step is **not** the backend yet. First put the prototype in the hands of 20 real users in one city for two weeks (section 2). That shows which screens people really use, checks the calculators against professionals, and collects the real prices and units the database will need. Then build the backend (Convex) around what was learned. Start with buyers and vendors, then add workers, engineers and delivery.

1. Fix first (before testing with real users)

Fix	Status	Why
Make Nearby Hire and Apply work.	Done	Testers would think the app is broken.
Upgrade to Expo SDK 57.	Done	Fixes a known memory problem.
Stop the keyboard covering forms; fix Checkout typing.	Done	Most testers use Android.
Set the EAS preview build to make an installable APK; project passes expo-doctor (21 of 21 checks).	Done	Testers can install from a link.
Create a "My jobs" screen or point "View all" to an existing one.	To do	Opens a "not found" page today.
Save the chosen profile on the phone and add a "My dashboard" button on the Profile tab.	To do	Testers should not pick their role every time. Needs only on-phone storage, no backend.
Make the remaining dead buttons work, or hide them for the test (list in the first document, section 8).	To do	Each dead button costs tester trust and wastes feedback.
Show what testers create: placed orders in Orders, new listings in My Store, work listings in Professionals, hire requests on the worker side.	To do	Testers need to see the result of their action to judge the flow. Can use an in-app store like the engineer one.
Keep test data on the phone between app restarts (cart, store, projects, saved places).	To do	A two-week test spans many app restarts.
Link the Delivery dashboard or hide it.	To do	Unfinished screens confuse reviewers.
Hide the admin entry in test builds.	To do	The PIN "1234" is inside the app.
Add a "Send feedback" button (WhatsApp or a form) and crash reporting (for example Sentry).	To do	Collects feedback and crashes from the phones directly.

2. Test before building the backend

Changing a screen now costs hours; changing it after the database, payments and login are built costs weeks. This test answers three questions: **do people understand the flows**, **are the calculators right**, and **what real data must the backend hold**.

2.1 How to run the test

Item	Plan
Length and place	Two weeks, one city (Abuja or Lagos).
Testers (20)	7 buyers, 5 vendors, 5 workers, 3 engineers or quantity surveyors. At least 5 on low-cost Android phones (2 to 3 GB memory), a few older users, a few on weak networks.
Install	Android first. Run *npx eas-cli build -p android --profile preview*. The *preview* profile in eas.json already builds an **APK**, so testers install it straight from the build link (no Play Store). Answer *Yes* to "Generate a new Android Keystore?" on the first build. iPhone testers later through TestFlight (needs an Apple Developer account).
Brief	A short WhatsApp message: this is a test app, the data is sample data, nothing real is bought or paid, and how to send feedback.
Group	One WhatsApp group per profile for questions, plus a 15-minute call with each tester at the end.

2.2 Tasks for each profile

Ask each tester to try these without help. Write down if they finish, how long it takes, and where they hesitate.

Profile	Task	Success when
Buyer	Find Dangote cement using Search.	Opens the product in under 30 seconds.
Buyer	Scan a printed sample barcode (section 2.4).	Product opens on the first or second try.
Buyer	Use the Cement calculator for a 10 × 8 m floor, 100 mm thick, and add the bags to the cart.	Cart shows the bags (the app gives 54 with 5% extra).
Buyer	Check out with Pickup station.	Reaches the success screen without help.
Buyer	Find an available mason near you and send a hire request for 3 days.	Card shows "Requested".
Vendor	Create a listing with 2 photos, price and stock.	Published in under 3 minutes.
Vendor	Restock a low-stock item in My Store, then mark a listing as sold.	Stock and status change.
Vendor	Turn off "Accepting orders" in Settings; save the shop location with GPS and rename it.	Both done without help.
Worker	Create a work listing.	Published; no field hidden by the keyboard.
Worker	From the dashboard, find jobs for your trade and apply to one.	Button shows "Applied".
Engineer	Create a project, log an expense, schedule a site visit and mark it done.	All four done; numbers on the dashboard change.
Engineer	Move a spec from Draft to Approved; call a team member.	Status changes; phone dialler opens.

2.3 Phone checklist

Tick each item on at least one low-cost Android phone and, when available, one iPhone.

Check	How
Keyboard	Type in every form (Add Listing, Create Work, Checkout, Saved Locations, calculators). No field or button may hide behind the keyboard.
Permissions	Deny camera, gallery and location the first time, then allow them. The app must explain and recover, never crash.
Barcode scanner	Scan the printed codes below in good and poor light, with and without the flashlight.
GPS	Saved Locations, Checkout "Use my current location" and Professionals "Near me", indoors and outdoors. Check the address is correct.
Dropdowns	Trade, Sort, Category and Project-status menus must open directly under their button.
Status bar	After leaving a green category screen, the clock and battery must be readable on light screens.
Back button	Every screen can be left with the back arrow and the Android back gesture.
Large text	Set the phone to large font. Text must not be cut off or overlap.
Weak network	Use the app on slow data and in airplane mode. It must not crash (it needs no internet yet, except maps and directions).
Small screen	Use a 5-inch phone. Bottom buttons must stay reachable.

2.4 Sample barcodes for the scanner test

Print these EAN-13 codes with any free online barcode generator and stick them on a box or sheet of paper.

Product	Barcode	Product	Barcode
Dangote Cement 42.5R	6151234000015	Ceramic Floor Tiles 60×60	6151234000053
Elephant Cement	6151234000022	PVC Water Pipe 3/4"	6151234000060
Aluminium Roofing Sheet	6151234000039	Sandcrete Block 9"	6151234000077
Iron Rod 12 mm	6151234000046	Emulsion Paint 20 L	6151234000084

A QR code with the text *buildlinka://product/p1* (p1 to p8) also opens the product.

2.5 Check the calculators with professionals

Give a professional the same example, compare their figure with the app, and aim to be within 10%. If not, the assumption is one line in the app's *lib* folder and is easy to change.

Calculator	Check with	Example (app result)
Cement	Mason or site engineer	10 × 8 m floor, 100 mm, 1:2:4, +5% → 54 bags, 5.9 t sand, 11.1 t granite
Steel & Iron	Structural engineer's bar schedule	5 × 4 m slab, 12 mm at 200 mm, top and bottom, +10% → 387 kg, 37 bars of 12 m
Roofing	Roofer's quote	12 × 10 m house, gable, 25°, 0.6 m overhang, aluminium, +10% → 100 sheets
Blocks	Mason	40 m of wall, 3 m high, 2 doors, 4 windows, 9", +5% → 1,073 blocks
Tiles	Tiler	5 × 4 m floor, 60 × 60, straight → 22 m², 62 tiles, 4 bags adhesive
Paint	Painter and the paint maker's data sheet	4 × 4 m room, 3 m high, 1 door, 1 window, 2 coats → 8.9 L
Electrical	Electrician or solar installer	6 bulbs, 2 fans, TV, 2 laptops, fridge (590 W), 4 h backup → 1 kVA, 2.5 mm² cable, 3.3 kWh battery
Plumbing	Plumber	5 people, 100 L each, 2 days → 1,000 L tank

2.6 Collect the real data the backend will need

Data	Why
Real product lists from 5 vendors: brands, sizes, units (bag, tonne, length, m², box, trip) and prices.	Becomes the first product catalogue and shows which units the database must support.
Delivery areas and fees from vendors and truck owners.	Needed to price delivery by distance and weight.
Day rates per trade and area from workers.	Realistic pricing and search filters.
How testers want to log in (phone code, email) and pay (transfer, card, USSD, pay on delivery).	Decides login and payment set-up.
The 5 most confusing screens and the 5 most wanted missing features.	Sets the priorities for the backend.
Whether vendors would pay a commission or a monthly plan, and how much.	Confirms the business model (section 10).

2.7 How to collect feedback

- A "Send feedback" button that opens WhatsApp or a short form, so feedback arrives with the screen name.
- Screen recordings, with the tester's permission, for the 15-minute end call.
- One shared bug list with three levels: **P0** crash or blocks a task, **P1** confusing, **P2** nice to have.
- Crash reports from the test build (for example Sentry) so crashes are seen even when no one reports them.

2.8 Ready to start the backend when…

- There are **no open P0 bugs**.
- At least **8 of 10 testers per profile** finish their tasks without help.
- Calculators are **within 10%** of the professionals' figures.
- A real product catalogue exists for at least 2 categories (for example cement and blocks).
- Login and payment choices are decided.

Tip: keep the app's screens the same during the test. Collect requests, then change everything once, before the backend, rather than shipping a new build every few days.

3. The backend (after testing)

Accounts and login

- **Phone number + SMS code** as the main login (most users are phone-first); email optional. Use a Nigerian-friendly SMS provider (for example Termii) or Twilio.
- **One account, several roles**: a vendor may also buy; an engineer may hire workers.
- Roles and permissions checked on the **server**, never only in the app. Replace the admin PIN with a real admin role.

Server and database: Convex

- **Convex** (the chosen backend): database, server functions, file storage and live updates in one place. Live updates suit order status, stock levels and project changes. Start on the free cloud plan; data can be exported and moved to a self-hosted Convex server later if needed.
- Product photos, profile photos and documents compressed on the phone, then stored in Convex file storage.
- Real **push notifications** (expo-notifications) for new orders, hire requests, applications, delivery updates, low stock and site-visit reminders.
- Works on slow networks: cache lists, show saved data offline, queue actions and retry when the network returns.
- **Photo search**: a server function that receives the photo (the app already sends it to *EXPO_PUBLIC_IMAGE_SEARCH_URL*), asks an image AI model, and returns matching products. The AI key stays on the server.

What the database must store (taken from the screens)

Area	Records
People	Users, roles, verification documents, saved locations, settings (accepting orders, alerts, hide profile).
Selling	Stores, products (with barcode, unit, photos, category), stock levels and movements, price history.
Buying	Carts, orders, order items, delivery method and address, payments, refunds.
Work	Work listings, hire requests (days, cost, status), job posts, applications, reviews.
Projects	Projects, members, expenses, site visits, specifications and their approval history.
Platform	Notifications, featured banners, reports, disputes, feedback, calculator results saved by users.

4. Payments and trust

- **Paystack** for card, bank transfer and USSD. Payment confirmed by the server (Paystack webhook), never only by the app.
- **Escrow / held payment**: BuildLinka holds the money and releases it to the vendor after delivery is confirmed, or to the worker after each job stage. This is the main reason people will trust the platform.
- Paystack split payments / subaccounts to pay vendors and workers automatically and keep BuildLinka's commission.
- Clear refund, cancellation and dispute process, with an admin screen to resolve disputes.
- Reviews only after a completed order or job, so ratings are real.

5. Suggestions per profile

Vendor

Suggestion	Real-world reason
Business verification (CAC registration, owner ID, shop address and photos) and a "Verified vendor" badge.	Buyers fear fake sellers.
Stock goes down automatically when an order is placed; vendor accepts or rejects each order.	Stops selling items that are out of stock.
Real units and bulk price tiers (half trailer, trailer load). Bring back bulk deals once the cart can apply discounts.	Builders buy in large, mixed units.
Fast price update screen (many prices at once).	Cement and rod prices change often.
Quote requests from calculators: a buyer's calculator result goes to several vendors who reply with a price.	Turns the calculators into sales for items not listed yet.
Delivery areas and fees; payout history and downloadable sales reports.	Heavy goods; vendors need records.

Worker

Suggestion	Real-world reason
Identity check (NIN), phone check, optional trade certificate, "Verified" badge.	Clients let workers into their property.
Portfolio of past work photos.	Clients judge quality by past jobs.
Accept, decline or counter-offer hire requests (price, date).	Real jobs are negotiated.
Stage payments held in escrow; availability calendar instead of only a switch.	Protects both sides; avoids double booking.
Phone numbers hidden until a job is accepted (in-app chat or call).	Keeps deals on the platform and protects privacy.

Engineer

Suggestion	Real-world reason
Verify the COREN licence before showing the badge.	Engineer approval is a legal and safety matter.
Shared project workspace for owner, engineer, workers and vendors.	One place for everyone on a site.
Bill of Quantities from the calculators (cement, rebar, blocks, roofing) that the buyer can order in one tap.	Connects planning directly to sales.
Site inspection reports with photos, GPS and time stamp, and a sign-off.	Proof of work and quality control.
Document upload with versions; budget "spent" filled automatically from real orders and worker payments.	Specs change; no manual typing of costs.

Buyer

Suggestion	Real-world reason
Order tracking with live status and a delivery code or photo on delivery.	Proof the goods arrived.
Compare the same product across vendors by price, distance and rating (started on the Cement screen).	Buyers shop around.
Save calculator results to a project; plan vs actual budget per project.	Most builds go over budget.
Turn a Build Project plan into a ready cart and a list of workers to hire.	Makes the planner useful, not just informative.
Schedule delivery date and time; pay in stages for large projects.	Sites cannot store everything; buyers build as money comes in.

6. Delivery (logistics)

- Add a fifth profile, **Driver / Logistics partner**, using the delivery dashboard that already exists.
- Delivery fee by distance and weight or volume (a truck of blocks is not a bike parcel). The calculators already give weights and quantities.
- Driver app: accept delivery, navigation, pickup confirmation at the vendor, delivery confirmation with the buyer's code and photo.
- Start by partnering with existing truck owners and logistics companies rather than owning vehicles.

7. Admin and operations

- Move the admin panel to a **separate web dashboard** with its own secure login (two-step verification), not inside the phone app.
- Approval queue for new vendors, workers, engineers and listings (the "under review" messages already expect this).
- Dispute and refund handling, user suspension, and reports of fake listings.
- Customer support by WhatsApp and phone, with the help center inside the app.
- Crash and error reporting and usage analytics to see where users drop off (start during the test, section 2).

8. AI features

- Connect the Building Assistant chat to a real AI model through BuildLinka's server; never put an AI key inside the app.
- Use the same tested formulas as the calculators and **real vendor prices** from the platform, so plans match the market.
- Photo search through the server (section 3).
- Always label results as estimates and offer "Get an engineer to review this plan", which also brings work to engineers.

9. Local needs, law and safety

- Nigeria Data Protection Act (NDPA): privacy policy, consent, secure storage of IDs and personal data.
- Terms of service covering payments, escrow, cancellations and disputes.
- Languages: English first, then Hausa, Yoruba, Igbo and Pidgin.
- Low-cost Android phones and slow networks: small images, fast loading, low data use; support large text.
- Payment options Nigerians use: bank transfer, USSD and card, and possibly pay on delivery for trusted buyers.
- Calculator results are estimates: keep the footnotes, and advise users to confirm with a professional before ordering large quantities.

10. How BuildLinka can earn money

Source	How
Commission	A small percentage on each material order and each worker job paid through the app.
Delivery margin	Part of each delivery fee.
Vendor subscription	Monthly plan for more listings, analytics and store branding.
Featured listings	Vendors and workers pay to appear in featured banners and at the top of search.
Quote requests	Vendors pay a small fee or commission for quote requests won from calculator results.
Pro tools	Paid project workspace features for engineers and construction companies.

11. Suggested roadmap

Phase	What to build	Goal
1. Fix and test (now, about 4 weeks)	The "to do" fixes in section 1, a test build, two weeks of testing with 20 users (section 2), then one round of changes.	Flows proven with real people; real data collected.
2. MVP launch in one city	Login, Convex backend, vendor listings and stock, buyer orders, Paystack with escrow, push notifications, admin approval. Vendor and Buyer profiles only.	First real orders.
3. Workers	Verification, hire requests and applications, stage payments, reviews, availability calendar.	Hiring through the app.
4. Engineers and delivery	COREN check, project workspace, Bill of Quantities to cart, site reports; driver profile and delivery tracking.	A whole project on one platform.
5. Growth	Real AI planner and photo search, quote requests, more cities, languages, subscriptions and analytics.	Scale.

Advice: test first, then launch the buyer–vendor order loop and make it excellent. When buyers trust BuildLinka with material orders, adding workers, engineers and delivery becomes much easier.